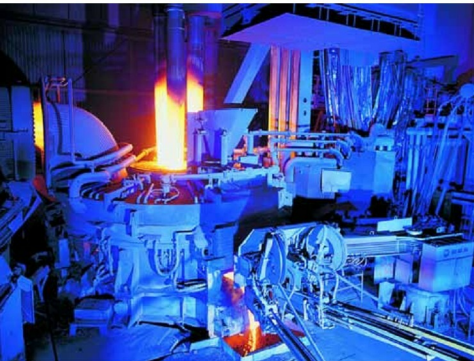


Industrial Automation v2

Fitting The Factory Into The Enterprise

Equipped with wireless and mobile connectivity, standards and platforms, artificial intelligence and machine learning, Web services and service-oriented architecture, etc, industrial automation has at last connected the factory to the rest of the enterprise. Not only has the factory been encapsulated within PCs and mobiles, it has also linked up with the business to respond to market situations, customer needs, etc in a timely and efficient manner



■ JANANI GOPALAKRISHNAN VIKRAM

PepsiAmericas, Pfizer, Conoco Phillips, Huntsman, Valero Energy, Sweeney Refineries and a clique of other global companies have deployed a fleet of 'grey collar' employees who practically remote-control the companies' business. Stephanie Neil (in *DeepDive: Enterprise Mobility - Let's Get to Work*) describes the grey collar crew as the union of the blue and the white

collared, brought together by mobility. It comprises executives on the road, engineers in the field and operators on the factory floor. They collect data on a constant basis as they go about their daily work, and feed it into the system.

The real-time information they generate has the power to control everything from logistics and marketing right down to production. That does not mean every employee has to give instructions to the various systems in the company, although they could do

so in extraordinary situations. On a regular day, they would simply have to update the system with information collected on-the-go. The information is promptly analysed and several dependent decisions are automatically taken and executed—how long should each machine work, how much of the produce should be packed, where should the inventory be, at what time

Some components of industrial automation

1. Artificial neural networks
2. Distributed control systems
3. Human-machine interfaces
4. Lab information management systems
5. Programmable automation controllers (PACs)
6. Programmable logic controllers (PLCs)
7. Supervisory control and data acquisition systems
8. Batch control systems
9. Simulation systems
10. Instrumentation
11. Motors and drives
12. Sensor networks
13. Industrial Ethernet and Fieldbus standards for robust communications
14. Wireless communication technologies
15. Platforms and middleware for software
16. Enterprise software
17. Software tools like Web services, workflow engines, etc